Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/14/2010
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Center of Acute Radiation Research		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-02 Center of Acute Radiation Research (CARR)
Start Date:	09/01/2008	**End Date:**	08/31/2013
No. of Post Docs:	5	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: title change per NSBRI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Freund, Gregory (University of Illinois at Urbana-Champaign) Ware, Jeffrey (University of Pennsylvania) Wan, X. (University of Pennsylvania) Weissman, Drew (University of Pennsylvania) Cengel, Keith M.D., Ph.D. (University of Pennsylvania) King, Gregory (Armed Forces Radiobiology Research Institute)		
Grant/Contract No.:	NCC 9-58-RE01801		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In this research, animal model systems are being utilized to evaluate the space radiation induced acute radiation syndrome (ARS) with vomiting (and/or retching) and white blood cell counts in ferrets, white blood cell counts, fatigue and immune system parameters in mice, and skin injury, with accompanying immune system changes, and white blood cell counts in pigs as the biological endpoints. In addition, the effects of combined exposure to simulated hypogravity and space radiation on blood cell counts and immune functions with respect to both the innate immune system and the acquired immune system in mice are being evaluated in mice. The first few years of the project are focused on determinations of whether certain parts of the ARS occur in animals exposed to space radiation(s) simulating that occurring during a solar particle event (SPE). If problems are observed to exist for the model systems used, later years of the project will be focused on the development of countermeasures for these effects. In these studies, selected drugs will be evaluated in the animal model systems to determine their effectiveness in prevention and/or mitigation of the specific ARS symptoms. The dose response relationships measured for these endpoints will be compared mathematically with published results for human subjects to determine the RBEs and the predictive value of these animal models for bio-dosimetry and for intervention studies to evaluate FDA-approved drugs that can be used for prevention and/or mitigation of the ARS symptoms. Various tissues will be collected from the animals for histopathological examination and for studies by various immunohistochemistry, qPCR and/or immunochemistry techniques to study the underlying mechanisms for the acute radiation syndrome and its prevention and/or treatment by drugs. These investigations have been designed to address Acute Radiation Syndromes - Specific Gaps 1, 2, 3, 5, 6, 7 and 8 identified in NSBRI-RFA-08-02.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	It is well-known that exposure to ionizing radiation at sufficiently high doses can result in various types of adverse biological effects. The countermeasures to be developed as part of this research program have significance for all individuals exposed to high doses of ionizing radiation on earth as well as in space. The countermeasures shown to protect against radiation induced biological effects from space radiation will be equally useful for protection against radiation induced biological effects on earth.		

Task Progress:	• Hematology : 1) Perform dose-response studies utilizing SPE-like protons and gamma radiation administered at high dose rate (HDR) and low dose-rate (LDR) to calculate relative biological effectiveness (RBE) values for hematopoietic effects. The effects of 70 MeV protons and gamma rays are comparable when administered at HDR or LDR, with RBE values of approximately 1. 2) Evaluate two animal models of reduced weightbearing, the Partial Weight Suspension (PWS) and hindlimb suspension (HLS) systems, in irradiated mice. Radiation exposure, with or without HLS or PWS, caused a significant decline in the WBC, neutrophil and lymphocyte counts. • Pig Skin: 1) Establish single fraction dose with SPE-like electrons to achieve acute grade 3 skin toxicity using adult mini-pigs. Results demonstrate acute grade 3 skin toxicity in animals receiving skin doses from 12.5-25 Gy. 2) Evaluate degree of change in hematopoietic and immune responses at each skin dose for up to 30 days post-irradiation. Acute immunologic changes observed and characterized for doses > or = 7.5 Gy. Acute hematologic changes observed in animals receiving > or = 7.5 Gy using more deeply penetrating (6+12MeV) but not superficial electrons. 3) Evaluate histologic evidence of skin compromise at each skin electron dose. Pathologic skin changes seen at all doses > or = 5 Gy, disruption of epithelial integrity and function seen at doses > or = 12.5. • Fatigue: 1) Explore 3 models of fatigue with gamma rays and protons at HDR and LDR. LDR, but not HDR, gamma ray and proton exposures lead to increased social withdrawal, but do not impact unmotivated locomotor activity or exercise tolerance. 2) Define RBE for proton vs photon induced fatigue. These results are seen with a threshold effect, where once a dose sufficient to produce a response is given, additional dose does not increase the magnitude of response. The lowest dose tested (50 cGy) was sufficient to produce effects with both gamma rays and protons, thus the RBE is currently undistinguishable from unity. • Immune System: 1) Increased levels of circulating LPS were found 24 hours after 2 Gy of SPE proton or gamma radiation delivered at HDR or LDR. A 2 Gy dose of SPE proton or gamma radiation induces the acute phase reactant LPS binding protein. Both types of radiation lead to bacterial translocation across the GI tract into the circulation, which then leads to activation of an acute phase response. SPE proton or gamma radiation induce a transient increase in circulating proinflammatory cytokines. 2) There is no evidence that these radiations induce a new immune response. • Ferret Vomiting/Retching: 1) Determined that prodromal responses to 0.25 Gy irradiations were similar to controls. 2) Preliminary data (6/group) show no apparent RBE difference for HDR irradiations between gamma-photons and 110 MeV protons, 3) No apparent dose-sparing effect between HDR and LDR 110-MeV protons at doses thus far evaluated.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
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